T20F256 FPGA CORE / MINIMUM 8 MOTOR CHANNELS, 12-CHANNEL PIN CAPACITY CANDIDATE

Exact 256 balls: Efinix T20-PINOUT worksheet T20F256. Pin allocation is a board candidate, NOT vendor fit or routed timing.
All VCCIO=3.3V. All VCC/VCC4=1.2V. No 5V-tolerant FPGA connection. No native LVDS: REF_RES intentionally NC.

One 100nF per supply ball: bulk per physical supply group. Decoupling is a layout-study allowance, not a PDN proof.
Active SPI x1 strap(C11: CS/TEST high; Flash bus reserved from user RTL. CC6 damping33R is provisional.
CDONE is open-drain with10k pullup; status does not grant motion permission. Configuration clock is internal, not T1.

U1
T20F256A / CANDIDATE PIN ALLOCATION

FPGA_CDONE	31	CDONE	
FPGA_CRESET_N	32	CRESET_N	
FPGA_FLASH_CS_N	33	GPIOL_00_SS_N	
FPGA_FLASH_CCK_SOURCE	34	GPIOL_01_CCK	
CDONE_FPGA	35	GPIOL_02	
FPGA_CC1	36	GPIOL_04_CS1	
	37	GPIOL_05_CS0	
FPGA_FLASH_MOSI	38	GPIOL_06_CS3	
FPGA_FLASH_MISO	39	GPIOL_09_CS5	
FPGA_CRESET_N	40	CC4	
FPGA_FREQ_TO	41	TO	
FPGA_FREQ_TDO	42	TDO	
FPGA_FREQ_TMS	43	TMS	
DONE_FPGA	44		
	45	GPIOL_12	
	46	GPIOL_13_CS2	
	47	GPIOL_14_CS3	
DONE_FPGA	48	GPIOL_15	
DONE_FPGA	49	GPIOL_16	
DONE_FPGA	50	GPIOL_18	
DONE_FPGA	51	GPIOL_19	
	52	GPIOL_20	
	53	GPIOL_20_CF04_D_CS4	
	54	GPIOL_21_CF04_D_CS5	
DONE_FPGA	55	GPIOL_22_CF04_D	
DONE_FPGA	56	GPIOL_23_CF04_D	
FPGA_CC_CDONE	57	GPIOL_24_CS45	
DONE_FPGA	58	GPIOL_25_CS45	
DONE_FPGA	59	GPIOL_26_CS45	
DONE_FPGA	60	GPIOL_27_CS45	
DONE_FPGA	61	GPIOL_28_CS45	
DONE_FPGA	62	GPIOL_29_CS45	
DONE_FPGA	63	GPIOL_30_CS45	
	64	GPIOL_31_CS47_CS46	
	65	GPIOL_32_CF04_A_CS47	
DONE_FPGA	66	GPIOL_33_CF04_D	
DONE_FPGA	67	GPIOL_34_CF04_D	
DONE_FPGA	68	GPIOL_35_CF04_D	
DONE_FPGA	69	GPIOL_36	
DONE_FPGA	70	GPIOL_37	
DONE_FPGA	71	GPIOL_39	
	72	GPIOL_40_CS48	
	73	GPIOL_41_CS48	
DONE_FPGA	74	GPIOL_42	
DONE_FPGA	75	GPIOL_43	
DONE_FPGA	76	GPIOL_44	
	77	GPIOL_45	
	78	GPIOL_46	
	79	GPIOL_47	
	80	GPIOL_48	
FPGA_USER_RESET_N	81	GPIOL_49	
PA_WATCHDOG_PA_N	82	GPIOL_50	
FPGA_ENDPOINT_READY	83	GPIOL_51	
FPGA_MOTION_REQUEST	84	GPIOL_52	
	85	GPIOL_53_CS50	
	86	GPIOL_54_CS51	
FPGA_PROGRESS_WD	87	GPIOL_55	
TX_POINT	88	GPIOL_56	
FPGA_J4M_FAULT	89	GPIOL_57	
FPGA_WATCHDOG_FAULT	90	GPIOL_58	
FPGA_DRIVE_FAULT	91	GPIOL_59	
FPGA_THERMAL_FAULT	92	GPIOL_60	
ENC1_A_FPGA	93	GPIOL_61	
ENC1_B_FPGA	94	GPIOL_62	
ENC2_A_FPGA	95	GPIOL_63	
ENC2_B_FPGA	96	GPIOL_64	
ENC3_A_FPGA	97	GPIOL_65	
	98	GPIOL_66_CDS12	
	99	GPIOL_67_CDS13	
ENC3_Z_FPGA	100	GPIOL_68	
FPGA_CTL_CBS11	101	GPIOL_69_CBS11	
FPGA_CTL_CBS10	102	GPIOL_70_CBS10	
	103	GPIOL_71_BOOT25	
	104	GPIOL_72_CF07B_CDS15	
	105	GPIOL_73_CF07B_CDS15	
ENC4_A_FPGA	106	GPIOL_74_PCLK0	
ENC4_B_FPGA	107	GPIOL_75_PCLK1	
ENC4_Z_FPGA	108	GPIOR_76_PCLK2	
ENC5_A_FPGA	109	GPIOR_77_PCLK3	
	110	GPIOR_78_CF07B_CDS31	
	111	GPIOR_79_CF07B_CDS30	
ENC5_B_FPGA	112	GPIOR_80	
ENC5_Z_FPGA	113	GPIOR_81_MBIST_EN	
HOME1_FPGA	114	GPIOR_82	
HOME2_FPGA	115	GPIOR_83	
HOME3_FPGA	116	GPIOR_84	
HOME4_FPGA	117	GPIOR_85	
HOME5_FPGA	118	GPIOR_86	
	119	GPIOR_87_CDS29	
	120	GPIOR_88_CDS28	
HOME6_FPGA	121	GPIOR_89	
HOME7_FPGA	122	GPIOR_232	
HOME8_FPGA	123	GPIOR_233	
	124	GPIOR_234_CDS25	
	125	GPIOR_235_CDS24	
HOME9_FPGA	126	GPIOR_236	
HOME10_FPGA	127	GPIOR_90	
HOME11_FPGA	128	GPIOR_91	
HOME12_FPGA	129	GPIOR_92	
ALM1_FPGA	130	GPIOR_93	
	131	GPIOR_94_CS27	
	132	GPIOR_95_CS26	
ALM2_FPGA	133	GPIOR_96	
ALM3_FPGA	134	GPIOR_97	
ALM4_FPGA	135	GPIOR_98	
ALM5_FPGA	136	GPIOR_99	
ALM6_FPGA	137	GPIOR_100	
	138	GPIOR_101	
	139	GPIOR_102_CS23	
	140	GPIOR_103_CS22	
ALM7_FPGA	141	GPIOR_104_CF04_A	
ALM8_FPGA	142	GPIOR_105_CF04_A	
ALM9_FPGA	143	GPIOR_106_CF04_A	
ALM10_FPGA	144	GPIOR_107_CF04_A	
ALM11_FPGA	145	GPIOR_108_CF04_A	
PWRBE_FPGA	146	GPIOR_109_CF04_A	
	147	GPIOR_110_CF04_A	
	148	GPIOR_111_CF04_A	
	149	GPIOR_112_CF04_A	
	150	GPIOR_113_CF04_A	
	151	GPIOR_114_CF04_A	
	152	GPIOR_115_CF04_A	
	153	GPIOR_116_CF04_A	
	154	GPIOR_117_CF04_A	
	155	GPIOR_118_CF04_A	
	156	GPIOR_119_CF04_A	
	157	GPIOR_120_CF04_A	
	158	GPIOR_121_CF04_A	
	159	GPIOR_122_CF04_A	
	160	GPIOR_123_CF04_A	
	161	GPIOR_124_CF04_A	
	162	GPIOR_125_CF04_A	
	163	GPIOR_126_CF04_A	
	164	GPIOR_127_CF04_A	
	165	GPIOR_128_CF04_A	
	166	GPIOR_129_CF04_A	
	167	GPIOR_130_CF04_A	
	168	GPIOR_131_CF04_A	
	169	GPIOR_132_CF04_A	
	170	GPIOR_133_CF04_A	
	171	GPIOR_134_CF04_A	
	172	GPIOR_135_CF04_A	
	173	GPIOR_136_CF04_A	
	174	GPIOR_137_CF04_A	
	175	GPIOR_138_CF04_A	
	176	GPIOR_139_CF04_A	
	177	GPIOR_140_CF04_A	
	178	GPIOR_141_CF04_A	
	179	GPIOR_142_CF04_A	
	180	GPIOR_143_CF04_A	
	181	GPIOR_144_CF04_A	
	182	GPIOR_145_CF04_A	
	183	GPIOR_146_CF04_A	
	184	GPIOR_147_CF04_A	
	185	GPIOR_148_CF04_A	
	186	GPIOR_149_CF04_A	
	187	GPIOR_150_CF04_A	
	188	GPIOR_151_CF04_A	
	189	GPIOR_152	
	190	GPIOR_153	
	191	GPIOR_154	
	192	GPIOR_155	
	193	GPIOR_156	
	194	GPIOR_157_PLL4	
	195	GPIOR_158_CF04_A	
	196	GPIOR_159_CF04_A	
	197	GPIOR_160_CF04_A	
	198	GPIOR_161_CF04_A	
	199	GPIOR_162_CF04_A	
	200	GPIOR_163_CF04_A	
	201	GPIOR_164_CF04_A	
	202	GPIOR_165_CF04_A	
	203	GPIOR_166_CF04_A	
	204	GPIOR_167_CF04_A	
	205	GPIOR_168_CF04_A	
	206	GPIOR_169_CF04_A	
	207	GPIOR_170_CF04_A	
	208	GPIOR_171_CF04_A	
	209	GPIOR_172_CF04_A	
	210	GPIOR_173_CF04_A	
	211	GPIOR_174_CF04_A	
	212	GPIOR_175_CF04_A	
	213	GPIOR_176_CF04_A	
	214	GPIOR_177_CF04_A	
	215	GPIOR_178_CF04_A	
	216	GPIOR_179_CF04_A	
	217	GPIOR_180_CF04_A	
	218	GPIOR_181_CF04_A	
	219	GPIOR_182_CF04_A	
	220	GPIOR_183_CF04_A	
	221	GPIOR_184_CF04_A	
	222	GPIOR_185_CF04_A	
	223	GPIOR_186_CF04_A	
	224	GPIOR_187_CF04_A	
	225	GPIOR_188_CF04_A	
	226	GPIOR_189_CF04_A	
	227	GPIOR_190_CF04_A	
	228	GPIOR_191_CF04_A	
	229	GPIOR_192_CF04_A	
	230	GPIOR_193_CF04_A	
	231	GPIOR_194_CF04_A	
	232	GPIOR_195_CF04_A	
	233	GPIOR_196_CF04_A	
	234	GPIOR_197_CF04_A	
	235	GPIOR_198_CF04_A	
	236	GPIOR_199_CF04_A	
	237	GPIOR_200_CF04_A	
	238	GPIOR_201_CF04_A	
	239	GPIOR_202_CF04_A	
	240	GPIOR_203_CF04_A	
	241	GPIOR_204_CF04_A	
	242	GPIOR_205_CF04_A	
	243	GPIOR_206_CF04_A	
	244	GPIOR_207_CF04_A	
	245	GPIOR_208_CF04_A	
	246	GPIOR_209_CF04_A	
	247	GPIOR_210_CF04_A	
	248	GPIOR_211_CF04_A	
	249	GPIOR_212_CF04_A	
	250	GPIOR_213_CF04_A	
	251	GPIOR_214_CF04_A	
	252	GPIOR_215_CF04_A	
	253	GPIOR_216_CF04_A	
	254	GPIOR_217_CF04_A	
	255	GPIOR_218_CF04_A	
	256	GPIOR_219_CF04_A	
	257	GPIOR_220_CF04_A	
	258	GPIOR_221_CF04_A	
	259	GPIOR_222_CF04_A	
	260	GPIOR_223_CF04_A	
	261	GPIOR_224_CF04_A	
	262	GPIOR_225_CF04_A	
	263	GPIOR_226_CF04_A	
	264	GPIOR_227_CF04_A	
	265	GPIOR_228_CF04_A	
	266	GPIOR_229_CF04_A	
	267	GPIOR_230_CF04_A	
	268	GPIOR_231_CF04_A	
	269	GPIOR_232_CF04_A	
	270	GPIOR_233_CF04_A	
	271	GPIOR_234_CF04_A	
	272	GPIOR_235_CF04_A	
	273	GPIOR_236_CF04_A	
	274	GPIOR_237_CF04_A	
	275	GPIOR_238_CF04_A	
	276	GPIOR_239_CF04_A	
	277	GPIOR_240_CF04_A	
	278	GPIOR_241_CF04_A	
	279	GPIOR_242_CF04_A	
	280	GPIOR_243_CF04_A	
	281	GPIOR_244_CF04_A	
	282	GPIOR_245_CF04_A	
	283	GPIOR_246_CF04_A	
	284	GPIOR_247_CF04_A	
	285	GPIOR_248_CF04_A	
	286	GPIOR_249_CF04_A	
	287	GPIOR_250_CF04_A	
	288	GPIOR_251_CF04_A	
	289	GPIOR_252_CF04_A	
	290	GPIOR_253_CF04_A	
	291	GPIOR_254_CF04_A	
	292	GPIOR_255_CF04_A	
	293	VCCIO4_3A	
	294	VCCIO4_3B	
	295	VCCIO4_3C	
	296	VCCIO4_3D	
	297	VCCIO3_3A	
	298	VCCIO3_3B	
	299	VCCIO3_3C	
	300	VCCIO3_3D	
	301	VCCIO3_3E	
	302	VCCIO3_3F	
	303	VCCIO3_3G	
	304	VCCIO3_3H	
	305	VCCIO2_3A	
	306	VCCIO2_3B	
	307	VCCIO2_3C	
	308	VCCIO2_3D	
	309	VCCIO2_3E	
	310	VCCIO2_3F	
	311	VCCIO2_3G	
	312	VCCIO2_3H	
	313	VCCIO1_3A	
	314	VCCIO1_3B	
	315	VCCIO1_3C	
	316	VCCIO1_3D	
	317	VCCIO1_3E	
	318	VCCIO1_3F	
	319	VCCIO1_3G	
	320	VCCIO1_3H	
	321	VCCIO0_3A	
	322	VCCIO0_3B	
	323	VCCIO0_3C	
	324	VCCIO0_3D	
	325	VCCIO0_3E	
	326	VCCIO0_3F	
	327	VCCIO0_3G	
	328	VCCIO0_3H	
	329	VCCA_1A	
	330	VCCA_1B	
	331	VCCA_1C	
	332	VCCA_1D	
	333	VCCA_2A	
	334	VCCA_2B	
	335	VCCA_2C	
	336	VCCA_2D	
	337	VCC_1A	
	338	VCC_1B	
	339	VCC_1C	
	340	VCC_1D	
	341	VCC_1E	
	342	VCC_1F	
	343	VCC_1G	
	344	VCC_1H	
	345	VCC_2A	
	346	VCC_2B	
	347	VCC_2C	
	348	VCC_2D	
	349	VCC_2E	
	350	VCC_2F	
	351	VCC_2G	
	352	VCC_2H	

U2
W25Q32JVSSIQ / 4 V, SPI

U3
ASE-50.000MHZ-LC-T

U4
74LVC1G17GW / CDONE DELAY SCHMITT

U5
74LVC1G32GW / USER RESET AND

U6
74LVC1G32GW / USER RESET AND

J1–J15 ARE INTERNAL LOGIC BOUNDARIES, NOT FIELD TERMINALS. Enables and extra home/alarm/encoder channels are reserves only.
CRESETN/74 plus two internal pullups screened. The separate 33k/1nF CDONE reset delay is a candidate, not a guaranteed time or safety function; threshold/partial-power confirmation required.

J1
POWER / QUALIFIED SEQUENCER

CORE_1V2	1
VL_3V3	2
LOGIC_GND	3
FPGA_CRESET_N	4
FPGA_LOCAL_RESET_N	5
FPGA_CDONE	6
FPGA_CTL_RESETN	7
LOGIC_GND	8

J2
JTAG SERVICE / NO POWER INJECTION

VL_3V3	1
LOGIC_GND	2
FPGA_TMS_TCK	3
FPGA_TMS_TMS	4
FPGA_TMS_TDI	5
FPGA_TMS_TDO	6
FPGA_CDONE	7
LOGIC_GND	8

J3
HOST SPI / DIO ONLY

HOST_SPI_CS_N	1
HOST_SPI_SCLK	2
HOST_SPI_MOSI	3
HOST_SPI_MISO	4
LOGIC_GND	5
FPGA_WATCHDOG_FAULT	6
FPGA_WATCHDOG_FAULT	7
FPGA_THERMAL_FAULT	8

J4
HW PERMISSION / NO ESP MOTION AUTHORITY

PA_WATCHDOG_PA_N	1
FPGA_ENDPOINT_READY	2
FPGA_MOTION_REQUEST	3
FPGA_PROGRESS_WD	4
TX_POINT	5
FPGA_J4M_FAULT	6
FPGA_LOCAL_RESET_N	7
LOGIC_GND	8

J5
TWELVE MOTOR STEP / DIR

STEP0_FPGA	1
DIR0_FPGA	2
STEP1_FPGA	3
DIR1_FPGA	4
STEP2_FPGA	5
DIR2_FPGA	6
STEP3_FPGA	7
DIR3_FPGA	8
STEP4_FPGA	9
DIR4_FPGA	10
STEP5_FPGA	11
DIR5_FPGA	12
STEP6_FPGA	13
DIR6_FPGA	14
STEP7_FPGA	15
DIR7_FPGA	16
STEP8_FPGA	17
DIR8_FPGA	18
STEP9_FPGA	19
DIR9_FPGA	20
STEP10_FPGA	21
DIR10_FPGA	22
STEP11_FPGA	23
DIR11_FPGA	24

J6
ENABLE RESERVES / CIRCUITS ABSENT

DRIVE_ENABLE1_FPGA_RESERVED	1
DRIVE_ENABLE2_FPGA_RESERVED	2
DRIVE_ENABLE3_FPGA_RESERVED	3
DRIVE_ENABLE4_FPGA_RESERVED	4
DRIVE_ENABLE5_FPGA_RESERVED	5
DRIVE_ENABLE6_FPGA_RESERVED	6
DRIVE_ENABLE7_FPGA_RESERVED	7
DRIVE_ENABLE8_FPGA_RESERVED	8
DRIVE_ENABLE9_FPGA_RESERVED	9
DRIVE_ENABLE10_FPGA_RESERVED	10
DRIVE_ENABLE11_FPGA_RESERVED	11
DRIVE_ENABLE12_FPGA_RESERVED	12

J7
HOME / 5–5 PRESENT; 6–11 RESERVED

HOME1_FPGA	1
HOME2_FPGA	2
HOME3_FPGA	3
HOME4_FPGA	4
HOME5_FPGA	5
HOME6_FPGA	6
HOME7_FPGA	7
HOME8_FPGA	8
HOME9_FPGA	9
HOME10_FPGA	10
HOME11_FPGA	11
HOME12_FPGA	12

J8
ALARM / 1–5 PRESENT; 6–11 RESERVED

ALM1_FPGA	1
ALM2_FPGA	2
ALM3_FPGA	3
ALM4_FPGA	4
ALM5_FPGA	5
ALM6_FPGA	6
ALM7_FPGA	7
ALM8_FPGA	8
ALM9_FPGA	9
ALM10_FPGA	10
ALM11_FPGA	11
ALM12_FPGA	12

J9
ENCODERS / 0–1 PRESENT; 2–5 RESERVED

ENC1_A_FPGA	1
ENC1_B_FPGA	2
ENC2_A_FPGA	3
ENC2_B_FPGA	4
ENC3_A_FPGA	5
ENC3_B_FPGA	6
ENC4_A_FPGA	7
ENC4_B_FPGA	8
ENC5_A_FPGA	9
ENC5_B_FPGA	10
ENC6_A_FPGA	11
ENC6_B_FPGA	12

J10
AUX / EXISTING ISOLATED INPUTS

PWRBE_FPGA	1
TOOLSTICK_FPGA	2
DOOR_STATUS_FPGA	3
PROTECTIVE_STATUS_FPGA	4
SPINDLE_FAULT_FPGA	5
SPINDLE_READY_FPGA	6
TOOL_HOLD_FPGA	7
COOLANT_FPGA	8
LUBE_STATUS_FPGA	9
SPARE_FPGA	10
VL_3V3	11
LOGIC_GND	12

IP5305
SUPPLIES 5V LIFESTEAK

SUPPLIES 3V LIFESTEAK

RESET: U3 combines qualified upstream reset with service RUN for configuration reset only. U4/U5 make a separate CDONE-qualified user-logic reset; R22/C39 provide a candidate post-CDONE delay and service RUN can hold it low through a JTAG handoff. No programmer output fights the sequencer.
User-logic reset release threshold, RC tolerance, discharge time, tUSER handoff and startup/release synchronizers remain bench/HDL qualification gates.
J2 is a logical service boundary; actual **4PLUG** connector, adapter, cable loading and ESD are unresolved.
JTAG programming requires a deliberate **user+cpu+pwm**, valid rails and machine outputs inhibited. No hot-plug or phantom powering.

NC crosses are genuine unused GPIO/config outputs and REF_RES only; every supply/GND ball is connected.
Boundary symbols are not released connectors. No footprint approval, PCB, DRC, bitstream, qualification or fabrication files.

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Title:

Size: A0 | Date: | Rev:
KiCad E.D.A. 10.0.4 | Id: 1/1